

Antiproliferative activity of lignans against the breast carcinoma cell lines MCF 7 and BT 20.



PURPOSE: Phytoestrogens are plant-derived, non-steroidal phytochemicals with anticarcinogenic potential. The major structural classes are the isoflavones and lignans. The aim of this study was to compare the effect of the plant-derived lignans secoisolariciresinol and matairesinol with the human lignans enterodiol and enterolactone as well as with 17β estradiol and tamoxifen on cell proliferation of breast carcinoma cell lines. **METHODS:** The influence of the lignans, 17β estradiol and tamoxifen on cell proliferation was determined using the BrdU test in MCF 7 and BT 20 cell lines. **RESULTS:** Enterodiol and enterolactone induced a stronger inhibition of cell growth in MCF 7 and BT 20 cells than secoisolariciresinol and matairesinol. The inhibition effects were less expressed in the BT 20 than in the MCF 7 cells. **CONCLUSIONS:** The human lignans enterodiol and enterolactone are more biologically active than their precursors secoisolariciresinol and matairesinol, and may be defined as the real drugs in cancer prevention.